

Staging design principles

The aim of this coursework is to investigate the optimal staging configuration of the **Minotaur IV+** showed in fig.1, a four-stage launcher, which is the enhanced version of the standard one, the Minotaur IV. The upgrade is represented by the more powerful Star-48BV upper stage instead of the Orion-38 used by the standard configuration.

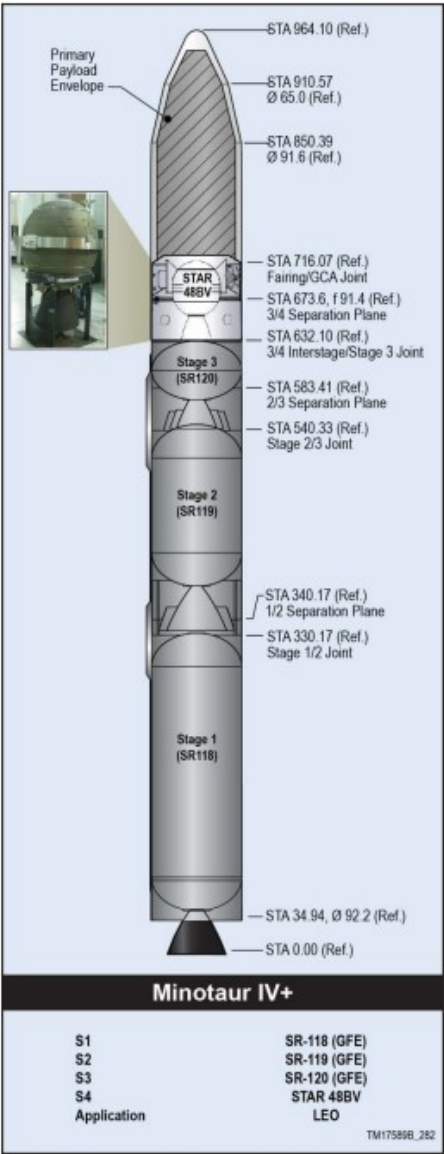


Figure 1: Minotaur IV+ schematics

Step 1: velocity increment

1.1 Total ΔV of the launch vehicle

In order to evaluate the total velocity increment of the launcher, one must consider the equations for the multistage rockets.

As expected, the evaluation is started from the analysis of the fourth sub-rocket because the given payload mass coincides with the payload mass of the last sub-rocket and the same it is valid for the overall mass of the launcher, which is evaluated from the sum of the given propellant mass, total inert mass and payload mass. Then one can evaluate the structural efficiency, mass ratio and ΔV from Tsiolkovsky law. The computations continue for all the below sub-rockets, remembering that the payload mass of the i -th subrocket is the same as the overall mass of the $i+1$ sub-rocket. The most important equations are:

$$m_{0_i} = m_{p_i} + m_{s_i} + m_{pl_i} \implies MR_i = \frac{m_{0_i} - m_{p_i}}{m_{0_i}} \implies \Delta V = -c_i \ln MR_i \implies \lambda_i = \frac{m_{pl_i}}{m_{0_i}}$$

The results in terms of mass budget and ΔV partitioning of the launcher are given in table 2.

sub-rocket	1	2	3	4	overall
$\Delta V[\text{m/s}]$	2006.82	2932.92	2718.80	1997	9655.54
$\Delta V[\%]$	20.78	30.38	28.16	20.68	
λ	0.446	0.298	0.341	0.459	0.0233
k_s	0.0735	0.116	0.0841	0.0716	-
MR	0.487	0.379	0.379	0.498	-

Table 1: ΔV partitioning of the Minotaur IV+ configuration

1.2 Time evolution of the launch vehicle mass

The mass time-evolution can be obtained from the definition of mass flow rate:

$$\dot{m} = -\frac{dm}{dt}$$

and assuming that it remains constant during the flight as the specific impulse of the SRM and consequently the thrust, one can integrate the previous expression from the zero time entering the given values for each stage and obtaining:

$$m_i(t) = m_{0_i} - \frac{m_{p_i}}{t_{b_i}} t \quad \text{where} \quad \dot{m}_i = \frac{m_{p_i}}{t_{b_i}} \quad (1)$$

The result is shown in fig.2. One can notice that the time evolution of the launcher mass has a linear decreasing trend, as shown by the time dependence of the equation 1. The dotted lines correspond to the burnout time of each stage. Here there are discontinuities related to the discarding of the inert mass m_{s_i} of each stage that take place when stage separation occurs.

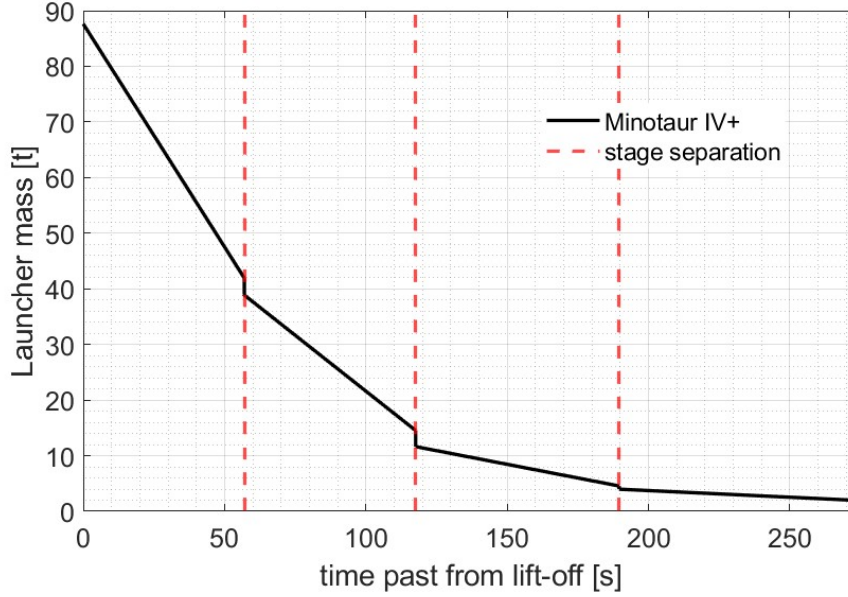


Figure 2: Lunch vehicle mass time history

1.3 Time evolution of the launch vehicle acceleration

The time evolution of the acceleration is obtained from the second Newton law applied at each stage:

$$a_i(t) = \frac{F_i}{m_i(t)} = \frac{\dot{m}_p c}{m_i(t)} = \frac{m_{p_i}}{t_{b_i}} \frac{c_i}{m_i(t)} \quad (2)$$

where $m_i(t)$ is the time evolution of the launcher mass obtained in the previous section.

The result is shown in fig.3. In this case there is an increasing trend of the acceleration during the burning time of each stage, due to the fact that the total mass of the launcher is decreasing. The increasing of acceleration at each stage is dictated by the relationship between initial acceleration, final acceleration and mass ratio of each stage:

$$\frac{a_{0_i}}{a_{f_i}} = MR_i$$

so, a stage with a higher mass ratio corresponds to a smaller difference between initial and final acceleration. Moreover, at stage separation time there is a very large discontinuity, which corresponds to the difference between the final acceleration of one stage and the initial acceleration of the next stage.

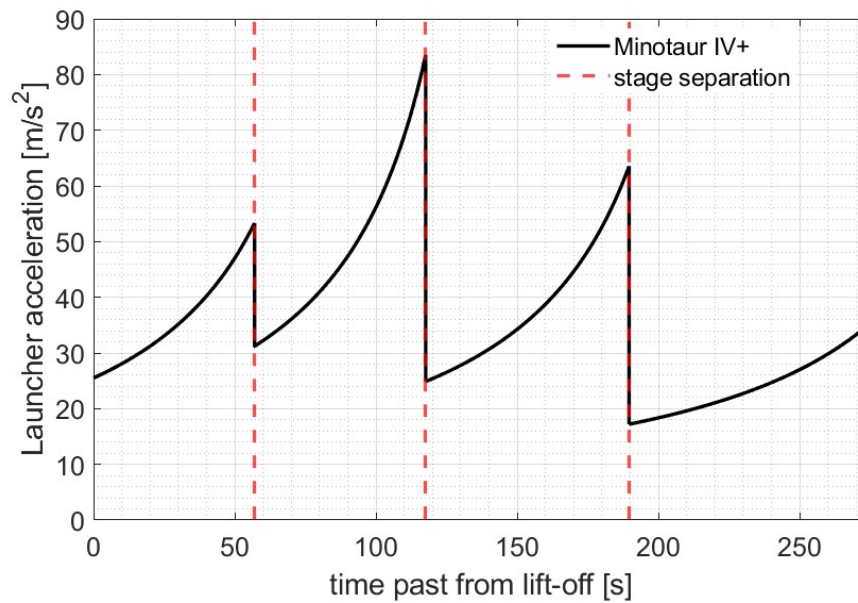


Figure 3: Launch vehicle acceleration time history

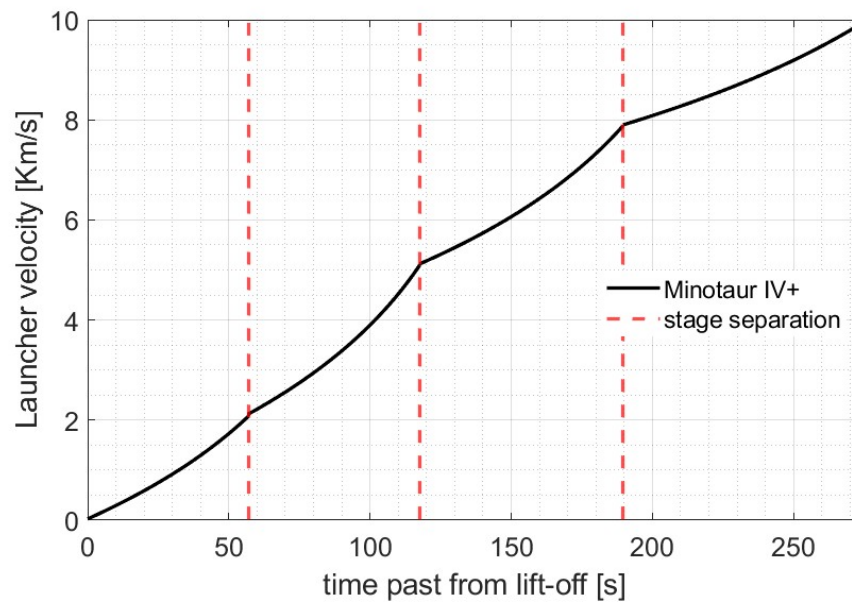


Figure 4: Launch vehicle velocity time history

1.4 Time evolution of the launch vehicle velocity

From the Tsiolkovsky equation it is easy to find that the instantaneous velocity at time t of each stage is:

$$V_i(t) = V_{0_i} - c_i \ln \left(\frac{m_i(t)}{m_{0_i}} \right) \quad \text{where} \quad V_{0_{i+1}} = V_{0_i} + \Delta V_i \quad \text{being} \quad V_{0_1} = 0 \text{ m/s}$$

The result is shown in fig.4. One can notice that the velocity has an increasing trend without discontinuities because in the argument of the natural logarithm in the Tsiolkovsky equation appears the mass ratio and in particular the initial mass of the stage m_{0_i} which corresponds to the final mass of the previous stage. The increase of velocity given by each stage is related to its mass ratio, in particular, an higher mass ratio corresponds to a lower velocity increase and vice-versa because a larger mass must be accelerated with the same thrust.

Step 2: optimal staging

The aim of the second step is that of finding the optimal staging configuration, fixing the stage technology (i.e. c and k_s) and the mission requirements (i.e. the total ΔV and the payload mass). This is done because the original feature of this launcher was a three-stages intercontinental ballistic missile, whose architecture was optimized for suborbital flight, while the Minotaur has an additional stage with a LEO mission requirement.

This is done adopting a parametric approach: the ΔV partitioning is varied considering different cases and then each case is plotted in order to catch graphically the most performing staging architecture that minimizes the lift-off mass m_0 in order to obtain the maximum total payload ratio λ_{tot} . Despite the simplicity of this method and its not very high precision, it can still be useful for a first qualitative analysis of the launcher's behaviour in the optimal configuration, with the awareness that for more reliable results an analytical approach with constrained optimization is necessary.

In fig.5 one can evaluate the obtained results. The most important one is that the optimized configuration (in green) is different from the actual ΔV partitioning (in black), as expected. Moreover, one can notice that the equal partitioning configuration is not the optimum one because the four analyzed stages are not identical in terms of specific impulse and structural efficiency, but despite this, it is very close to the optimized configuration. This is due to the fact that the specific impulses and k_s are close to each others and the obtained result does not differ so much from the equal partitioning case.

The optimized configuration is obtained varying both $\Delta V_2/\Delta V_{tot}$ and $\Delta V_3/\Delta V_{tot}$. The numerical results are shown in table 2. Comparing them with the results of the actual configuration in table 1 it is evident that:

- the λ_{tot} is decreased from 0.0233 of the actual configuration to 0.0214 of the optimized one
- it is convenient to decrease the contribution to ΔV_{tot} of the second and third stages in favour of the first and last one, which corresponds to an increase of the mass ratio of sub-rocket 2 and 3
- as a consequence of the previous point, there is an increase of propellant mass for stage 1 and 4 and a reduction for the two intermediate stages.

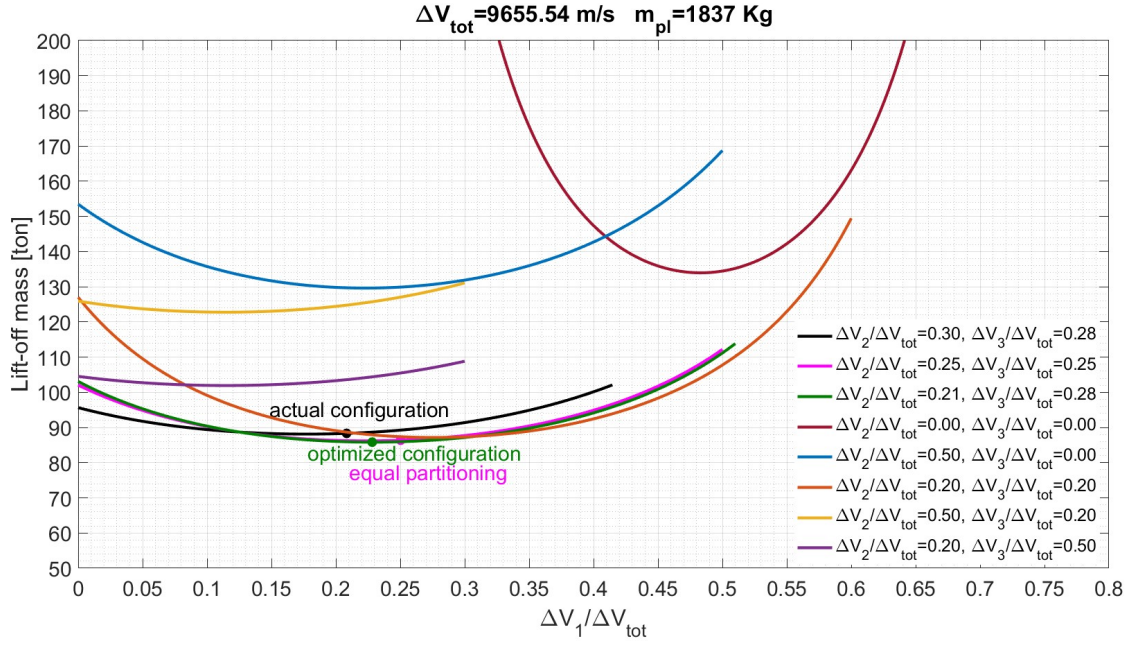


Figure 5: Parametric optimal ΔV partitioning: lift-off mass, m_0 , as a function of ΔV_1 , for different combinations of ΔV_2 and ΔV_3 .

sub-rocket	1	2	3	4	overall
ΔV [m/s]	2218.88	2048.58	2660.74	2727.34	9655.54
ΔV [%]	22.97	21.21	27.58	28.25	
λ	0.451	0.443	0.350	0.338	0.0214
MR	0.451	0.507	0.404	0.386	-
m_0 [Kg]	85989.3	35031.8	15524.6	5429.7	85989.307
m_p [Kg]	47213.7	17253.7	9246.1	3335.5	77048.875

Table 2: Optimized ΔV partitioning of the Minotaur IV+ configuration through parametric analysis.